

Scheduling and Rescheduling of Railway Operations: A Review and Expository Analysis

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Abstract Railway operations are planned in four levels of hierarchy: (i) strategic, (ii) tactical, (iii) operational control and (iv) real-time control. *Scheduling* is an initial time allocation of resources to meet demands in completing a task and *rescheduling* is a later modification of such resource allocations. An extensive set of some major railway scheduling and rescheduling operations as published in literature is analyzed in this paper. We observe that many authors consider scheduling as a strategic or tactical operation and rescheduling as an operational or real-time control. Several unexplored issues that are strategic or tactical per se, but hold considerable rationale for operational and real-time control (and vice versa) of railway operations are brought out. It is interpreted that (i) in practice, both scheduling and rescheduling are inadvertently performed at all hierarchical levels and (ii) operational feasibility at a hierarchical level is dependent on effectiveness of modeling at other hierarchical levels. Finally we present an array of potential research issues that are significant in practical railway operations. Our discussions are based on an understanding of objectives, modeling details, approaches and issues arising from the diversity in railway operations and to a less detail on solution methods and techniques.

Keywords Scheduling · Rescheduling · Train pathing · Timetabling · Line planning · Optimal resolution · Heuristics · Meta-heuristics · Crew planning · Policy decisions

Introduction: Operations Planning and Management

Operations in every commercial and service industry are planned, executed and managed under four broad hierarchies; they are (i) strategic, (ii) tactical, (iii) operational control and (iv) real-time control. The hierarchical differences are based on decision levels, time horizon of planning and execution, stakeholders, criticality and impact of decisions, and performance threats. The interrelation between the four is that operational feasibility at a hierarchical level is strongly influenced by modeling details at other hierarchy. The focus of this paper is on a key area of transportation research, i.e., railway operations. We analyze scheduling and rescheduling of railway operations at different hierarchies in this paper, as observed from research publications and practice in railway industry.

Railway operations are hierarchically classified based on several factors. Motivation behind strategic operations is two-fold: Superior class of public utility and profitability. Strategic operations are handled by railway policy makers; some examples are infrastructure development, reliability, energy efficiency, capacity planning, pricing and revenue decisions, type and size of service requirements, layout and route planning and interaction with other transport modes. Strategies are planned to be implemented in the order of few years and decision changes are initiated based on policy differences and developmental requirements. In many countries, railway operations are managed either by private service providers (operators) or by local divisional

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heads. Railway operators implement strategic decisions as annual operational plans and monitor the service quality and revenue handling. Some decisions at the tactical level are line planning, timetabling, routing, crew scheduling and rolling stock circulation, traffic, nodal and yard management. Tactical plans are developed annually, monthly, or weekly as applicable and changes are initiated on a need basis from the next lower hierarchy. Operational control is executed by section managers within each division. Time span at operational control is lesser than that of tactical level and operational control could be periodic (cyclic/repetitive), or aperiodic (non-repetitive) and planned (deterministic) or unplanned (stochastic). Some of the operations at operational control are shunting, rostering, corridor (platform) occupancy, rolling stock handling, signaling, periodic maintenance of infrastructure. Infeasibility at the tactical level affects service quality, but the available time to react and repair any potential disturbance is very small. Real-time control is at the lowest hierarchy of operations; labor employees are generally the first ones to drive real-time control operations. A real-time control is performed when events that are potential threats to operational criticality and safety arise as emergencies. These events are usually stochastic in nature and necessitate contingency resolutions. We illustrate in Fig. 1 the hierarchical classification of operations and the factors based on which an operation is grouped under a particular hierarchy. As observed in the figure, strategic and tactical operations are at the planning level and operational control and real-time operations are at the execution level.

At the outset, policies and procedures trickle down hierarchically from a planning level to an execution level. We interpret later that effectiveness at each hierarchy, including feasibility and preparedness to stochastic disturbances are strongly influenced by modeling details at higher hierarchies. Published literature on a wide range of railway operational issues that validate our interpretations is discussed in this paper.

The rest of the paper has four major sections. We discuss the similarities and differences between scheduling and rescheduling operations in terms of objectives, methods and

approaches in “[Scheduling and Rescheduling](#)” section. “[Railway Operations: An Overview of Terminology](#)” section is on railway operational terminology. In “[Strategic and Tactical Scheduling: Current Scenario](#)” section, some major planning level operations that fall under strategic railway management such as line planning, timetabling management, rolling stock management, crew planning and rostering and policy decisions are discussed. In “[Rescheduling Railway Operations](#)” section, some execution level operations, particularly transportation operations are elaborated. We summarize our insights briefly with details from key published literature and issues of practical relevance in both sections. “[Conclusive Remarks](#)” section has our conclusive remarks and interpretations of scheduling-rescheduling in railway operations management along with chronologically tabulated literature.

Scheduling and Rescheduling

Scheduling is a time allocation of resources to meet demands in completing a task. For deterministic or stochastic reasons (that are beyond the scope of this discussion), sometimes it is required to reallocate resources for task completion which is referred as *rescheduling*. Both scheduling and rescheduling are directed towards achieving specific objectives while maintaining feasibility of safety and preferential constraints. Some common scheduling objectives are cost minimization and maximization of customer satisfaction. The most common rescheduling objective is to recover the target initial schedule.

An extreme case of infeasibility in constraints satisfaction is a deadlock. Deadlock is a situation, wherein two or more tasks mutually demand resources allocated to each other and subsequently one of the tasks has to be aborted or retracted. If constructive heuristic procedures are used to sequence tasks one after another, then deadlocks can be prevented. Other computer-aided sequencing procedures are likely to induce deadlocks, which need to be detected and resolved.

Several authors (Assad (1980); Jovanovic and Harker (1991); Cai and Goh (1994); Carey and Lockwood (1995) and Higgins et al. (1996)) grouped scheduling as a strategic or tactical operation and rescheduling (Ko et al. (2008); Luèthi et al. (2007a); Kraay and Harker (1995)), as an operational or real-time control operation. Such a sharp divide is refuted in this paper by expository discussions of published literature. In most of the available literature, both scheduling and rescheduling when posed as mathematical programming problems are formulated very similarly, except for the objective function. Recovering the original schedule at each resource (say, at one station) and for each task (say, for a train) is unlikely in such rescheduling



Fig. 1 Operational hierarchies and the classification factors

models, which defies the basic purpose of rescheduling. Operationally the allowable time-frame to complete can vary and it is possible that rescheduling be allowed a large time-frame and scheduling be allowed a small time-frame. Further in rescheduling, if all tasks could not be feasibly rescheduled through all resources as in the original schedule, few services are cancelled in partial or in full or re-routed. Under these premises, we elucidate that a real-world railway operation should be addressed as scheduling or rescheduling based on the intended objective and be classified under a hierarchy based on the allowable time-frame and upon which the approaches and methodology to solve the problem should be decided. We classify several key railway operations as available in research literature under the four hierarchies and our classification is illustrated in Fig. 2. We report our analysis that led us to these remarks.

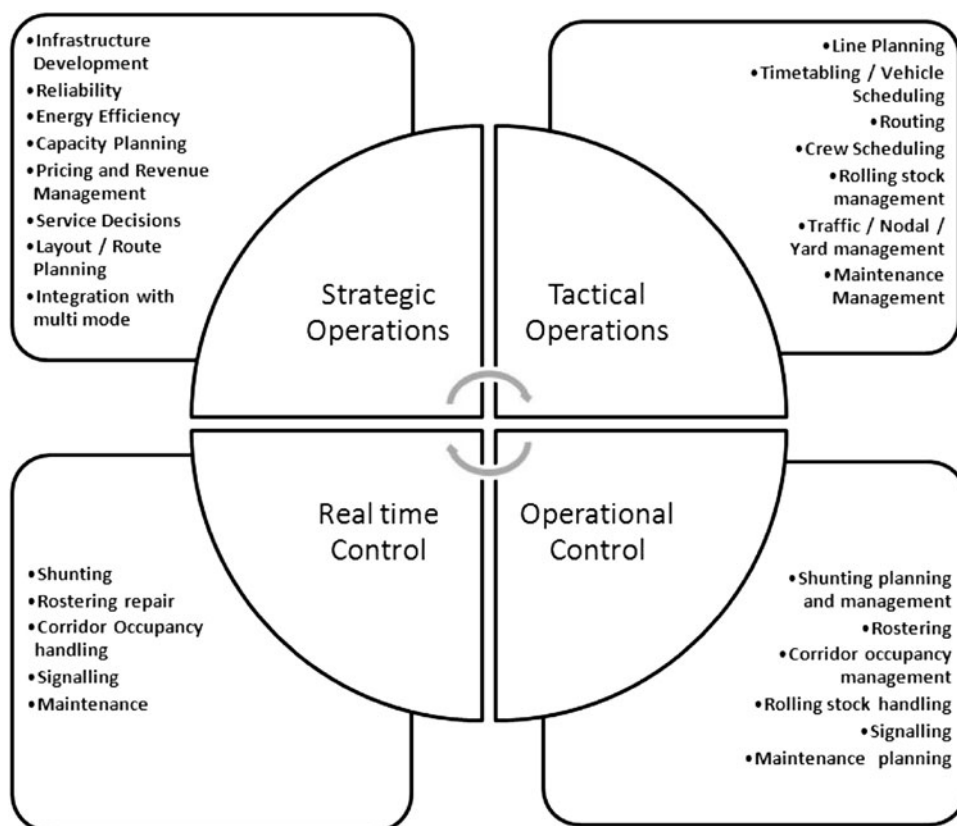
Transportation is the core railway operation that is continued to be researched by academicians and practitioners over several years. A wide variety of publications is available in transportation scheduling; though not very recent, detailed literature reviews are also available. In particular, Assad (1980), Cordeau et al. (1998) and Tornquist (2005, 2006) provide a detailed survey of railway literature, discussing transportation modeling, problem description, solution approach, control strategy used in

solution, layout descriptions, objectives and constraints, evaluation and testing. It should be noted here, that without strong exceptions scheduling operations are treated at the planning level and rescheduling operations are treated at the execution level in research writings. We provide a broader overview of several other planning and execution operations, as available in the literature. This operational classification is important to understand objectives, approaches in modeling, methodologies, solution techniques and inter-relations between the problems.

Railway Operations: An Overview of Terminology

Railway is a unique public transportation mode that offers as many management challenges as revenue and public utility. Railway is a highly favored transportation in many regions such as Americas, Europe, South Asia, North Africa, China and Australia, both in passenger and freight service. Management and co-ordination of railway operations are tedious as every single operation requires several types of resources—infrastructure, carriages (vehicle units), communication, personnel and all with dynamic status variations. Physical features and utility of these resources are widely varied; they are tailored to suit different services, such that sharing of certain resources

Fig. 2 Railway operations classified under four hierarchies



among different types of services is often infeasible. The interdependencies between these diverse resources are very complex and can increase in multi-folds as the operation size increases. Further to resources, policies that monitor railway operations are sometimes quite rigid to confirm regional regulations and to ensure safety and reliability as a public utility. In spite of management challenges, service benefits from railway operations are huge as railway is the only mass public transportation mode that can cater to widely distributed remote and difficult terrains. The fuel consumption is much to the advantage in comparison to other transportation modes and hence business interest in terms of revenue and public utility service is stupendous. There are several factors that determine the complexity of issues, effectiveness and efficiency of planning and management at the core of railway operations. Some of these vital factors are succinctly defined as follows.

Infrastructure: Rail Track, Rail Units, Stations, and Communication

Rail tracks between stations are laid as single or in multiples to support train movements in a single or bi-direction. Further track layout between stations is formed as a linear or network topology. Rail units comprise of engines and locomotives (carriages). There are different types of engines and locomotives for various service classes. A train consists of one or more engines and a combination of carriages, as per the operating route. Stations are halt locations that enable passenger and goods to board and off-load. Communication devices involved in railway operations are plenty; the main elements are the signaling equipment. The signaling equipment indicates halting and movement of trains or caution driving. Train running speed, acceleration, deceleration, and many other traffic characteristics are determined by rail infrastructure. More elaborate discussions on infrastructure and modeling infrastructure can be found in Rangaraj et al. (2003), Gokhale (2004), Chang et al. (1998) and Cai and Goh (1994).

Rolling Stock

Rolling stock refers to the physical parts of a train that is moved from one location to another as part of the transportation. Carriages and engines are the major railway rolling stock.

Service Categories

Railways cater to passenger and freight transportation. In many railway administrations, passenger traffic is precisely timetabled to be followed strictly. Freight traffic is not

timetabled and are operated when the tracks are idle. Passenger traffic is further sub-classified into two: (i) Long distance passenger traffic that supports follow-up and overtaking train movements and (ii) Sub-urban (metro) traffic that supports only follow-up train movements. Freight transportation problems are discussed in Kraay and Harker (1995), Salim and Cai (1997), Blum and Eskandarian (2002), Godwin et al. (2007a, b), and a passenger transportation problem is discussed in Isaai and Singh (2000). Schedule represented as a timetable offers easy readability to the general public, and schedule represented as arrival/dispatch sequences is a very useful document for railway personnel to handle traffic management. Schedules that are not timetabled as that of freights are represented in arrival and departure sequences at each station.

Spatial and Temporal Horizon

Space in railway research articles is specified by track configuration and length, in general. Single, double or multiple tracks are considered and tracks are laid in a linear or a network topology. Temporal specifications indicate the time interval of railway operational domains. Tornquist and Persson (2005) discussed a procedure to decide rolling time windows and to break down a fixed time duration scheduling problem into multiple flexible time duration problems to achieve improved performance objectives. Spatial and temporal factors determine the actual size of real-life problems captured in the model and hence determine the problem complexity.

Solution Approach

Several solution approaches are proposed, developed and deployed to resolve issues in railway operations. For a major variety of problems, optimal, heuristic, meta-heuristic, partially heuristic solutions and evolutionary approaches are proposed and studied. Interested readers are directed to elaborate review articles on railway operations, issues, and solution approaches by Davidsson et al. (2005), Tornquist (2006, 2007) and Caprara et al. (2007).

Solution Models: Objectives and Constraints

Performance objectives are defined to improvise and evaluate railway operations based on pre-defined criteria. Constraints are the operational conditions under which the system operates. Some constraints are hard and critical for operational safety and others are soft and are desirable to achieve customer satisfaction. Detailed survey of research in railway operations modeling is available in Assad (1980), Cordeau et al. (1998) and Caprara et al. (2002).

Policies and Regulations

Government policies determine railway operational rules and regulations. In many European countries, infrastructure is provided by the Government and private operators manage the commercial operations, earn profit after payments for infrastructure usage. In many Asian and American countries, both infrastructure management and traffic operators are a single body. Complexity of modeling is different in both, as the European model is predominantly competitive and the Asian model is co-operative. Some traffic is across International borders and follows trade and commerce guidelines between the nations. These apart, infrastructure, new and additional layouts and services, co-ordination between other transport modes, capacity decisions, pricing of various classes of service and managing revenue are other operations that are under the jurisdiction of Government policies.

Crew and Personnel

Railway personal are posted at the station locations or boarded on trains. Both traveling crew and crew at stations are allowed certain period of rest between not more than 8 h of duty.

Rosters and Timetables

Roster is a time-ordered list of tasks that are assigned to crew. From a roster, it is also possible to read tasks assigned to any crew member at a particular time. Timetables are time-ordered lists of arrivals and departures of trains at each station. Alternately, it is also possible to read the scheduled arrival and departure time of any train at each station on its route.

Discussions on scheduling and rescheduling in the next two sections follow the above terminology and definitions.

Strategic and Tactical Scheduling: Current Scenario

Unlike other transportation modes, complexities arising from interdependence and mix of activities abound in railways. All railway operations draw immense research interest, but a major of publications deal with tactical and operational control issues. Some of these are theoretic discussions and some present substantial details on implementation, testing and evaluation. Research papers on strategic and real-time control of railway operations are fewer compared to that of air traffic. The major of published railway operations are grouped according to Caprara et al. (2007) and D'Ariano and Pranzo (2009) as line planning, timetabling, platforming, rolling stock management,

shunting, and crew planning. As observed by Caprara et al. (2007), optimization techniques for long term capacity studies on infrastructure, rolling stock, crew and real-time models are not presented in any. In the subsections that follow, railway literature on (i) line planning, (ii) timetabling management, (iii) rolling stock management, (iv) crew planning (rostering) and (v) policy decisions are analyzed. Each sub-section summarizes typical problem descriptions and some interesting problem variations that are not dealt with earlier.

Line Planning

A line refers to a cyclic schedule of a set of trains in a particular route and direction of travel. Typically city traffic (metro) lines are planned for a 24 h duration and long distance passenger traffic lines are planned for anywhere between 24 h to a week. In few countries, freights are not timetabled and in many others freights are not scheduled in lines. The more common performance objectives that are modeled are (i) Maximizing the number of services and (ii) minimizing operational costs. One of the early optimization models in line planning is developed by Bussieck (1998). An integer linear programming model is developed for line planning to satisfy customer demands. A relaxation approach using branch and bound heuristic is also proposed. Goossens (2004) reported extensively on various models and algorithms to solve some line planning problems. Estimating passenger demand is one important criterion to line planning, and statistical and forecasting tools are required to make a realistic estimate of passenger demand. Lindner (2000) proposed a cost optimal model for line planning using branch and bound method. Carey and Lockwood (1995) proposed a strategic model and algorithm using branch and bound for pathing an additional train in an existing schedule. Other optimal, heuristic and meta-heuristic approaches to scheduling of railway traffic are reported in Jovanovic and Harker (1991), Higgins et al. (1996), Cai and Goh (1994), Chiang et al. (1998), Ghoseiri and Morshedsolouk (2006), Isaai and Singh (2000), Nachtigall and Voget (1996), Hüisman et al. (2005), Diaz (2001), and Dorfman and Medanic (2004); some of these articles are discussed more elaborately in the next section. Michaelis and Schöbel (2009) proposed re-ordering of the three operations—line planning, timetabling and vehicle scheduling with a view to improve customer orientation and minimize planning costs. Line planning is considered a strategic railway operation because of the longer planning phase. Modification in a planned line is necessitated for strategic reasons such as service quality improvement. Research attempts toward quality improvement are discussed in the rest of this subsection.

Energy Efficiency

Energy conservation is one of the recent developments in railway operations planning. As a public transportation, railways demand a huge energy. Ko et al. (2008) proposed to minimize energy consumption in railways by optimizing train run-curve using dynamic programming. The optimization procedure involves changing speed-position profile, while maintaining the same running time. Many optimal train run-curve optimization problems assume non-regenerative electric and mechanical braking, whereas model developed by Ko et al. (2008) is based on regenerative braking. This objective is more complex to achieve than a line scheduling problem, as the run-curve has to be optimized with runtime as maintained in the scheduled lines. Kawakami (1999) proposed a power saving approach in planning high speed railway.

Speed Profile Optimization

Train movements are modeled either with a fixed or a variable speed profile. Fixed speed profile assumes the runtime of a train is uniform and fixed in a segment and variable speed profile incorporates acceleration and deceleration near curved tracks and at halt points. Speed profile optimization is directed toward realistic operations modeling, and to improve efficiency of operations with respect to train dynamics. Fixed speed train profiles and extending time rolls are discussed in D'Ariano (2008). Another approach to voluntary train speed variations and driver behavior in optimizing conflict resolution objectives is discussed using a customized heuristics in Albrecht (2009). He proposed a heuristic to optimize driving speed and the proposed approach is evaluated against sequential dispatch procedure.

Capacity Maximization

Luèthi et al. (2007b) observed that capacity of a railway network may be increased by three means: (i) Infrastructure development, (ii) Reduction of block time using advanced signaling systems and (iii) Operational modifications by reducing the buffer time added to train traversal times. All of them are strategic decisions in railway transportation. They further discussed that buffer times are reserve times added to a train's halt time or travel time or headway (arrival or departure and meet). Headway buffer allows some time for dispatchers to react to disturbances and stabilize the schedule. Run time buffer reduces impact of runtime variations and train dynamics. Both types of buffers are essential to reduce knock-on delays, to prevent train pile-up and to eliminate deadlocks. Gokhale (2004) and Rangaraj et al. (2003) addressed some aspects of

capacity maximization using signaling systems as a simulation model. Minimizing trip time is discussed in Khan and Zhou (2009). A point of contention is that how are strategic objectives maintained when disruptions occur while the schedules are being deployed and are known just-in time. Some methods and models to overcome such a condition are briefed in Luèthi et al. (2007a) and Khan and Zhou (2009). Capacity maximization, specifically in busy networks can be very complicated. Improving the efficiency of a heavy (busy) network is elaborately analyzed in Luèthi (2009). Increasing capacity with additional infrastructure, optimal models that map capacity demands to additional resources, optimal, heuristic, meta-heuristic approaches to capacity maximization are some problems that are not addressed in research articles.

Reliability

Hüisman et al. (2005) mentioned punctuality and average train delay as two measures of reliability. Delays caused by emergency events are most difficult to handle, and require real-time approaches to resolve. There is a paucity of research articles on real-time scheduling in railways, in contrast to air traffic. Reduction in lateness and various delay handling approaches are discussed in a majority of the publications. Most of them propose an operational control approach; a strategic approach toward reliability is proposed in a few. A list of research articles that propose minimizing delay in rail transportation includes Gorman (1998), Gatto (2007), Mazzarello and Ottaviani (2007), D'Ariano and Pranzo (2009), Chang and Chung (2005), Sahin (1999), Cheng (1998), D'Ariano (2008), Beder et al. (2000), Chen et al. (2009), Brucker et al. (2005), Chang and Thia (1996), Chiu et al. (1996), Cicerone et al. (2008), Diaz et al. (1999), Frede et al. (2008), Gatto et al. (2009), Gely et al. (2009), Heilporn et al. (2008), Jespersen-Groth et al. (2009), Hellstrom et al. (1998), Tornquist and Davidsson (2002), Kizilisik (1999), Lee and Ghosh (2001), Liebchen and Stiller (2009), Li et al. (2007), Luèthi et al. (2007b), Makkinga and Metselaar (1998), Walker et al. (2005) and Caimi et al. (2009). Many of them are heuristic, meta-heuristic and partially heuristic approaches to recover a conflict free schedule, when a planned schedule is disrupted. Procedural details of some of these approaches are discussed in a later section. Invariably all proposed models resemble a planned maintenance, with known disruption occurrence. Dispatch orders of disrupted trains determine punctuality and delay minimization in a schedule, which standard dispatch heuristics may not achieve. Other delay management approaches propose using Integer programming variables to define dispatch orders as in Schachtebeck and Schöbel (2008), Liebchen et al. (2007) and Schöbel (2007).

The four issues discussed in this section are (i) energy efficiency, (ii) speed profile optimisation, (iii) capacity maximisation and (iv) reliability. All four imply track occupancy plan modification, which is rescheduling for strategic objectives. Energy efficiency is achieved by minimizing braking frequency and maintaining a uniform speed profile. Capacity can be increased by minimizing the slacks in runtime/halt time and reliability is achieved by increasing time slacks and maintaining an optimal speed profile. A precise optimization model for any one of the above problems can be difficult to formulate. Integrating two or more of any of the above objectives can be still more difficult to model and solve. Some unexplored research problems that hold relevance in practice are (i) an optimization model for any of the discussed objectives, (ii) integration of some of the contradicting objectives to build a more generic model. Lindner and Zimmermann (2005) observed that line plan is generated using optimal models and crew plans and rolling stock plans are subsequently drawn from the optimal schedule. Hence a proposed schedule requires redesigning and adaptation in order to meet requirements of other sub-problems.

Timetabling Management

Timetabling is the process of providing a definite departure and arrival time for a set of trains in a railway network. A timetable is a user-friendly one-place listing of all trains that halt at and travel through station, all stations on a train route and scheduled arrival/departure times at every station. It also indicates travel time between two scheduled station halts and halt time durations at a station. It essentially reports absolute values of arrival-departure times of all trains and at all stations and also arrival-departure orders of all trains at each station. Railways produce two types of timetables intended for passenger traffic—One is a cyclic timetable and the other is a non-cyclic timetable. Cyclic timetable caters mainly to suburban traffic, where travel times of trains are uniform and scheduled halt durations are fixed. These timetables are easier for the user to remember, such as departing from station X at a:b hours, there is a train every c minutes to station Y. Non-cyclic timetables are developed by travel operators for long distance passenger trains to achieve certain performance objectives. Each train travel conforms to a set of safety and preferential constraints. Every travel and halt time in a portion of the section has an additional allowance called as reserve time to accommodate changes in driving pattern, train dynamics and passenger's boarding and alight time. Some earlier approaches to timetable generation applied branch and bound algorithm (Jovanovic and Harker 1991) and greedy algorithm (Cai and Goh 1994). A recent overview of optimal models and algorithms that can extract

utility information from timetables is published in Müller-Hannemann (2004). Some discussions included in this paper are multi-criteria optimization and shortest path algorithms. One sequencing heuristic is proposed by Burdett and Kozan (2008) to create new timetables. Timetabling is considered as a job shop problem and a heuristic combination of insertion, backtracking, dynamic selection is applied to generate timetables. Liebchen and Möhring (2004) considered timetabling as a periodic event scheduling problem and by extending the problem to a mixed integer linear problem (MILP) they observed that other decisions such as network planning, line planning, vehicle scheduling can be integrated into a timetabling problem. A robust approach to timetabling by absorbing delays with a fast heuristic approach that uses Mixed integer linear programming and stochastic optimization is proposed by Fischetti et al. (2007). Three different variations of delay resistance and robustness in timetabling are proposed by Liebchen and Stiller (2009), Liebchen et al. (2007) and Michaelis and Schöbel (2009). Another approach to model and solve timetabling problem is published in Caprara et al. (2002). Frede et al. (2008) proposed a new approach to extract timetable information in the presence of delays. Cacchiani (2008) proposed an Integer linear programming model and a Lagrangian relaxation approach to solve some real world instances of typical timetabling problems, such as timetabling in a single line corridor, freight transportation using an existing timetable and train unit assignment. Table 2 is a summarized report of the extensive literature study performed on mathematical models, heuristic procedures and experimental studies in railway operations. Research on railway operations is classified based on problem type, infrastructure details, problem formulation, control and solution mechanisms, evaluation levels and problem instances and size. Table 1 is a guideline of the framework of Table 2.

Certain interesting and unexplored objectives in timetabling problems are discussed below.

Reserve Distribution

Additional time in planned travel and halt duration is referred as *slack* or *buffer*. In a railway section, whether the entire railway operations are managed by a single body or by various private operators, each train holds a priority in the section. This priority defines the cost of delaying that train or profit it can earn. High priority trains have very little buffer added to their travel/halt times. Lesser the buffer time, more difficult is to restore the schedule during disruptions. Since all high priority trains hold the least buffer, during disruptions, less priority trains are off balanced more easily. Hence robustness of timetabling depends on reserve times and such reserve times need to be

Table 1 Framework of research literature in transportation management

Problem Type Approach (Author and Year)	Control Centralised (C) Distributed (D) Localised (L)
Infrastructure Representation First Parenthesis: Line(L), Network(N) Second Parenthesis: Block details Third Parenthesis: Station details Single(S), Double(D), Tracks(N) Uni-(U), Bi-(B)Directional Infinity - (α)	Solution Mechanism
Problem Formulation Objectives Constraints Model	Evaluation Level 1. Conceptual Approach 2. Simulated Experiments with Artificial Data 3. Simulated Experiments with real Data 4. Field Experiments 5. Implemented (Deployed)
	Problem Instance and Size

optimally distributed over the section. Some interesting questions are to find out how much buffer will a train need to maintain timetable robustness and how could the buffer be distributed?

Compensation and Condensation Area

A part of the section that has busy traffic is called as a dense (or condensation) area and the part that has fewer traffic is called as sparse (or compensation) area. This definition has another time dimension as peak and lean hours. A practitioner's approach is to minimize travel buffer time and maximize halt buffer time in a condensation zone and the reverse at the compensation zone. (i) identifying an appropriate objective for generating timetables in dense/sparse locations during peak/lean hours? (ii) study of impact of travel and halt buffer times on optimal resource occupancy are some of the critical research issues.

Integration of Multiple Services

Inclusion of other service types such as freight and special service trains in an existing timetable is not discussed in many published research, other than Godwin et al. (2007a, b). Other possibilities include integrating slow, fast and super-fast services in a working timetable. But these are routine practices in every railway management, planned using a simulation approach. Conventional optimization, heuristic models and solution approaches have not been featured for service integration. Khan and Zhou (2009) discussed stochastic optimization and heuristic approximation to a robust double track timetabling problem.

Disruption Handling

When services are disrupted either due to planned maintenance or of stochastic failures, both arrival-departure times and the arrival-departure orders of all trains have to be modified. In general, re-dispatch orders follow standard sequencing rules, such as earliest finish first, first come first, minimum tardiness etc. A collection of varied approaches, models, and solution procedures is found in the literature. One of the crucial timetabling decisions is to decide on a rescheduling objective. A second difficult decision is communicating timetable change over the entire network and to all stakeholders. Research articles in railway operations do not feature these issues.

Rolling Stock Management

Rolling stock comprises of railway resources that are on the move. These types of resources incur large initial investments, but recurring expenses are generally on maintenance. Rolling stock management is planned at strategic or tactical railway operation levels. When train units travel from one station to another, they are also routed in the opposite direction as a part of regular rail traffic. This facilitates the availability of train units for the next cycle of travel without incurring additional costs. In an ideal schedule, rolling stock should not run empty (without utility) which is quite difficult to achieve in the case of freights. Rolling stock management is significantly different in a dense and a sparse network. When the number of rail units that departs from one station does not arrive at the same station in some periodicity, it is called as an off-balance.

Table 2 Models and solution approaches in railway transportation: existing literature

Approach	Model	Infrastructure representation	Objective	Constraints	Solution mechanism	Control	Evaluation level	Problem instance and size
Salim and Cai (1997)	MIP	(L) (B:BS) (S:B α)	Min waiting and stopping costs	Safety, Consistency Occupancy/ Preferential	Determine meet with GA	C	3	Line: 12 segments 14 trains
Ingolitto et al. (2004)	CSP	(L) (B:BS, UD) (S:B α)	Min average traversal time for each new train	Bounds, frequency stop over, exclusive headways and precedence	Determine visit order using CSP formulation back-tracking is used	C	3	Line: 65 segments 81 trains
Blum and Eskandarian (2002)		(L) (B:BS) (S:B α)	Maximize profit		Reserve tracks for trains highest profit using MAS, heuristics, GA, CPM	C	3	Line: 200 segments 64 trains
Isaai and Singh (2000)		(L) (B:BS) (S:BN)	Minimum waiting times	Safety, travel-time halt-time capacity of track and station	Determine train visiting orders by earliest time of resource release order	C	3	Line: 200 segments 64 trains
Lin and Hsu (1994)		(L) (B:BS,UD) (S:BN)	Min delay of sacrificed train when solving local conflict		Start with infeasible schedule Shift slots from earliest first conflict	C	3	Line: 102 stations 350 trains
Chiang et al. (1998)		(L) (B:US,UD) (S:BN)	Valid time-table		Repair non-conflict by repair methods and earliest first principle	C	3	Line: 102 stations 350 trains
Caprara et al. (2002)		(L) (B:US) (S:B α)	Min Travel-Time exceeding ideal run time		Modify orders by lagrangian relaxation subgradient optimization	H	3	Line: 16 or 48 stations 221 or 54 trains
Keaton (1989)	Linear 0 - 1 MIP		Min operating time and costs		Lagrangian relaxation	C	3	
Chang and Thia (1996)	Pro-totype		Regularity risk congestion		Predictive fuzzy control	C	1	
Chang and Chung (2005)		(L) (B:US) (S:US)	Minimum total time in system, travel time and schedule deviation	Travel/Halt headways, load factor train capacity	Decide visiting order of trains at stations using GA	C	3	Line: 30 station 100 trains
Parkes and Ungar (2001)	Linear 0 - 1 MIP	(L)	Maximum total value costs	Travel, Halt consistency headway precedence follow-up/meet	Myopic best response strategy	D	2	Line: 15 trains

Table 2 continued

Approach	Model	Infrastructure representation	Objective	Constraints	Solution mechanism	Control	Evaluation level	Problem instance and size
Carey and Lockwood (1995)	Linear MIP	(B:US) (S:Bz) (N) (B:BS) (S:Bz)	Minimum travel and waiting time costs	Travel/halt arr/dep, meet headways consistency and bounds	Decide visiting order of trains at loops and tracks to path next	C	2	Line: 10 stations 28 segments 10 trains
Yoko and Norio (2005)			Maximize customer satisfaction		Partly cancellation, slot change delay, route changes suggested	C	3	
Higgins et al. (1996)	Non-Linear MIP	(L) (B:BS) (S:BN)	Min weighted delay and costs	Arr, Dep, meet headways travel/halt time, consistency	Branch and bound	C	3	Line: 14 stations, 31 trains
Jovanovic and Harker (1991)	Non-Linear MIP	(L) (B:BS) (S:BD)	Min tardiness costs	Arr, Dep, meet headways travel/halt time, consistency	Branch and bound heuristics to decide orders	C	4	Line: 130 meet points, 200 trains
Diaz (2001)	Linear MIP	(N) (B:BS) (S:BU)	Max number passengers	Consistency, halt and travel headways	Backtracking and what-if procedures	C	3	Line: 130 85 services
Chiu et al. (1996)	CSP	(L) (B:BS) (S:BS)	Min largest delay per train	Speed, occupancy, station entry station exit, line/halt time, arr, dep, meet headways	Varying heuristics to decide orders on segments	C	3	Line: several stations, BS segments
Beder et al. (2000)		(N) (B:BS) (S:Bz)	Fault prevention		Co-ordinate atomic actions			
Kraay and Harker (1995)	Linear MIP	(N) (B:BS) (S:Bz)	Min tardiness costs	Arr, Dep, Meet headways travel/halt time, consistency heuristics lp model	Column generation finds bounds and local search	C	3	Line: 11 meet points 16 segments
Ping et al. (2001)	IP	(L) (B:UD) (S:Bz)	Min total delay	Arr, Dep, Meet Headways Travel/Halt Time	GA to decide visiting orders at segments, start times	C	3	Line: Double tracked segments, 14 stations 250 trains
Hellstrom et al. (1998)		(L)	Min tardiness costs		Branch and bound to decide orders, meets, overtakes	C	3	Line: 23 single track segments, 20 trains

Table 2 continued

Approach	Model	Infrastructure representation	Objective	Constraints	Solution mechanism	Control	Evaluation level	Problem instance and size
Sahin (1999)	MIP	(B:BS) (S:BD) (L) (B:BS) (S:BD)	Min Delay of 2 local conflict resolutions	Preferential avoidance of trivial solution	Look-ahead heuristic to decide effectiveness of 2 solutions	C	2	Line: 19 stations, 20 trains
Vernazza and Zunino (1990)		(N) (B:-) (S:-)	Most urgent conflicts dealt first		Trains bid capacity to local DC allocation on local urgency	L	3	Network
Iyer and Ghosh (1995)		(N) (B:-) (S:-)	Each train minimizes its total travel time		Each train requests for N-tracks negotiates for allocation	L	3	Network: 50 stations, 84 segments
Tornquist and Persson (2005)		(N) (B:N) (S:N)	Min total delay costs		Find train visiting orders using Branch & Bound (IP solver CPLEX)	C	3	Network: 130 stations 136 segments, 93 trains
Ho and Yeung (2001)		(N) (B:BN) (S:BN)	Minimum total weighted delay		Decide order of track usage using TS, GA, SA	C	2	
Cai and Goh (1994)	IP	(L) (B:BS) (S:BD)	Min Stop and waiting costs	Arr, Dep, meet headways travel/halt time, consistency	Greedy algorithm to change orders, schedules	C	2	Line: 10 segments, 99 trains
Makkinga and Metselaar (1998)	Pro-totype		Conflict detection			C	1	

Budai et al. (2009) presented a model to re-balance rolling stock during disruptions in a view to minimize off-balances. Maròti (2006) discussed rolling stock management at four operational hierarchies, maintenance routing and optimal ways of solving them. Other objectives and constraints of this problem include carriage assignment, minimizing idle runs, restrictions on number of carriages that can be pulled by an engine, and unavailability of carriages. Another associated problem with rolling stock management is shunting and it is the process of moving empty and idle carriages into a shunting area. In that context, Fioole et al. (2006) presented a model for rolling stock circulation, by combining and splitting train units and rolling stock. This model is formulated based on passenger demands and various objectives are evaluated with a Dutch railway section data. Nielsen and Maroti (2009) presented a general framework for handling disruptions in rolling-stock management. Disruptions, as unavailability of rolling stock is updated on an online tool and schedules are updated, progressively on a rolling time horizon. The system is tested on real data from Dutch Railways and solutions are evaluated as a trade-off between quality and quickness. Cacchiani et al. (2008) developed a model to recover rolling stock and to evaluate robustness of rolling stock schedules. An optimal approach is applied to assess the price of robustness in rolling stock management in a Dutch passenger railway. Impact of adequacy in rolling stock to maintain punctuality and quality of services in passenger railway is discussed in (Abbink et al. 2002). An OPL optimization model is presented to allocate rolling stock to train services and to effectively manage rolling stock capacity.

Two features that are unique with rolling stock are that (i) when a rolling stock is idle or on the run, few other resources are simultaneously occupied; for example, when a carriage is stationed at a corridor, the corridor becomes unavailable for traffic. If the corridor has to be made available for other traffic, the carriage has to be moved to the yard and for this; an engine and a driver are needed to be allocated. (ii) An empty run of rolling stock not only reduces profit, but also adds to revenue losses from multiple resources. Occupancy of multiple resource types and the subsequent additional revenue loss are deterrent to rolling stock circulation. Some research papers propose a solution approach for rolling stock circulation; however implications of resource occupancy and revenue loss are not modeled in a research perspective.

Crew Planning and Rostering

Crew scheduling is the process of assigning personnel to specific operations. Considering that one task may require many crew personnel for completion and one person may be assigned to multiple tasks, rostering is ordering of tasks to

be completed by a crew as well as ordering of crew who will complete a task. Conflict free crew schedules and rosters are developed as strategic and tactical schedules. The most common objective in crew scheduling research is to minimize number of crew, and a variety of optimal and heuristic models can be developed for this problem. Vaidyanathan et al. (2007) observed that crew planning is very different in North America from European crew planning and much is left to be developed as crew is manually planned and managed. They formulated crew assignment as an integer-programming problem and crew management as network-flow based crew optimization problem. A significant feature is the proposed FCFS algorithm for crew assignment by problem decomposition and constraint relaxation. Walker et al. (2005) proposed a combined model to recover a crew schedule and a timetable from a perturbed schedule in real-time. Kroon and Fischetti (2000) proposed additional constraints to handle driver assignments, so that punctuality is achieved. Constraints on variable assignments of duties are proposed so that utility of drivers is robust. Dynamic column generation, Lagrangian relaxation and heuristics are applied to solve the combined scheduling and driver assignments. Möller (2002) presented brief discussions on models and solution approaches to crew scheduling and enumerated the difficulties in solving crew management optimally. He presented a two-phase approach for crew management as crew rostering and crew scheduling. Abbink et al. (2007) presented decomposition of large crew scheduling problem into smaller instances to evaluate decomposition approaches, specific to Netherlands railways. Abbink et al. (2004) developed a crew scheduling project using OR methods. Caprara et al. (1997) presented a decomposition approach to solve crew scheduling. Veenturf et al. (2012) is claimed to be the only one on crew rescheduling, when disruptions necessitate changes in crew management. Bojovic and Milenkovic (2010) discussed optimal crew scheduling and presented an integer-programming model for crew rostering. Freling et al. (2000) proposed a branch-and bound heuristic approach to crew rostering and scheduling and found that the solutions were suboptimal, but quick. Jütte et al. (2011) presented the most recent publication on crew scheduling for freight operations in the Netherlands. Goncalves et al. (2000) presented a mathematical model and a fuzzy approach to crew timetabling without the rostering step. Raghuram et al. (1986) presented a general model to minimize crew on a particular section of the Indian Railways. Tatsushi et al. (2001) proposed Dantzig-Wolfe decomposition approach to improve convergence of column generation in crew scheduling. Wang et al. (2006) proposed an ant-colony optimization approach and Taïeb (2001) proposed a network flow approach to crew scheduling. It is observed in (Veenturf et al. 2012) that no other research on crew rescheduling is

published. We observe that a major of publications in crew scheduling are focused around few European groups; a few generic models on crew management in other regions are available, which we attempted to bring out.

The uniqueness of crew scheduling is that either crew resources are available or they are absent. Issues such as crew absence and lateness either with some prior notice or on an emergency are difficult to be modeled. Crew absence or lateness with prior notification falls under the category of operational control in crew scheduling and emergency absence is a real-time crew scheduling problem. In both cases, if ideal strategic plans are followed and no substitute crew is available, rescheduling is impossible. Number of substitute crew required in a section is an interesting problem to explore using forecasting and regression analysis.

Policy Decisions

Few research articles discuss policy issues in railway operations. In general, policy issues are too many and vary widely from one administration to another. Research articles that address railway policy matters are very few; an attempt is made to discuss them here as profound as could be, albeit any classification. Arslan (2009) proposed a competency assessment and evaluation framework for American railways. It is a practice in all American transportation, to involve consumers in assessing quality of public services. Passenger convenience is a prime indicator of service quality in American railway operations and Arslan (2009) proposed evaluating the service quality based on consumer feedback. These evaluations form the main inputs to strategic modifications in the next transportation fiscal policy. It is not uncommon to evaluate service quality based on customer feedback in practice of other transportation modes also. But a research perspective in this exercise opens a flood of improvement possibilities. Mahadevan (2007) proposed a model to achieve a global service intention in passenger railway operations as against local service intention, specific to Swiss Federal Railways using a graph model. Wüest et al. (2008) extended this model to achieve a sustained global service objective in real-time. When various governmental policies are finally applied, Bergkvist et al. (2005) proposed a behavioral model for railway stakeholders. They developed a multi-agents based simulator tool to decide actions of all stakeholders, and hence to generate cost effective strategies for rail operators. Leander and Lukaszewicz (2008) developed a model that achieves energy efficiency and reduction in power and costs in rail transportation. This model has four interfaces: (i) to handle aperiodic schedule, (ii) to handle cyclic schedules, (iii) real-time controller that minimizes energy and power, and (iv) drive at lowest energy when all safety constraints are met. This model operates on inputs (rolling stock, infrastructure and power utility data) to

provide heuristic decisions on energy efficient driving by computing power limitations, runtime margin and load shedding. Campbell (1996) proposed a model to solve issues specific to inter-modal traffic, considering reliability, asset utilization and to maximize expected marginal revenue by exploiting train priorities.

These articles address policy issues in railway transportation—they raise questions such as (i) what is a rationale for a railway policy change? (ii) how are requirements for policy change identified and measured? (iii) what determines effectiveness in a policy change? (iv) how can this effectiveness be sustained in real-time? (v) what are the motives of train operators? (vi) what is the role of train operators in maximizing their profit margin with changing policies? and (vii) what strategies should the government adopt to maximize their profits? Each question is responded with some commendable solution approaches. But several other variations in modeling and solution approaches could be explored. Expectedly it warrants that changes in policies are inevitable and subsequently changes in schedules are also warranted.

Summary of Inferences

Some strategic scheduling problems in railway operations are discussed. In each category of problems, certain key issues that are still unexplored in a research perspective, but extremely relevant in operational practice are identified. We observe that there is a lack of an integrative approach to address a combination of problems in railway research and published literature. It is note-worthy that a change in schedule at the strategic level is in-fact a reschedule, as the objective is to recover the original schedule, satisfying few additional policy constraints. This classification gains significance as approaches and solving methodologies differ for strategic and real-time control problems. However research publications define and address rescheduling as an operational control or as a real-time control. We ascertain that strategic motives are known to drive schedule modifications and our insights on rescheduling performed at strategic hierarchies are listed in brief statements below.

- Rescheduling is not necessarily an implication of disruptions and conflicts, but could also imply strategic policy changes.
- Rescheduling objectives have to be aligned with strategic goals.
- Computational time for rescheduling at strategic levels is not as critical a levy as in rescheduling at lower hierarchies.
- Strategic level issues are solved using optimal and pareto-optimal approaches and execution level issues are solved using heuristic and meta-heuristic approaches.

Rescheduling Railway Operations

Rescheduling in railway operations is considered as an operational control or a real-time operation in available literature. In an earlier section, we discussed about crew rescheduling (Walker et al. (2005), Potthoff (2010a, b) and Veelenturf et al. (2012)) and rolling stock rescheduling (Nielsen and Maroti (2009) and Budai et al. (2009)). A high number of publications are available in transportation rescheduling and we discuss the more significant ones in this section. Caprara et al. (2002) observed that rescheduling disrupted railway traffic has been modeled only as an operational control problem. As an operational control problem, it is presumed that disruption occurrence is known a little earlier than the actual instance of occurrence and while trains are at locations before the disruption location. Planned maintenance is a typical case of operational control rescheduling; whereas a real-time rescheduling refers to disruption handling when disruptions are known just-in time. A real-time and optimal rescheduling model is proposed in Narayanaswami (2009). Methodical discussions of optimal scheduling approaches that are applicable in rescheduling problems and operational control rescheduling approaches are discussed respectively in the two subsections that follow. A majority of them are heuristic, or partially heuristic or meta-heuristic approaches. Our interpretations and insights on schedule-reschedule incompatibility are presented as a section summary.

Optimal Approaches

Carey and Lockwood (1995) proposed one of the earliest and acclaimed models to train pathing on a single track railway. They developed a Mixed Integer Programming (MIP) model to minimize cost associated with train arrivals and departures. Follow-up and overtaking are the two permissible train movements in their proposed model. A heuristic solution is proposed to path any new service into the existing network with the objective of minimizing arrival/departure costs and to preserve the arrival/departure orders of the scheduled trains. This approach can be adopted to reschedule disrupted services, when disrupted trains can be re-instated into the existing time-table as new services. Another optimal model to schedule trains on a single track is proposed by Higgins et al. (1996) as a decision support tool to optimally dispatch trains in real time and as a planning tool to evaluate impacts of changes in time-table and infrastructure. A branch and bound procedure is applied to solve the model that decides dispatch priority to minimize the current delay and an estimate of remaining overtaking and crossing delay. Cai and Goh (1994) developed a MIP model for scheduling trains on a single track. It is observed that potential crossing conflicts

are hard to solve exactly and they proposed a basic greedy heuristic procedure to generate sub-optimal feasible solutions that can be employed for realistic applications. Kraay and Harker (1995) proposed an optimal model to schedule freights in real time and link the model with strategic solutions. The model is decomposed into an integer and a continuous variable sub-problem. The integer variable sub-problem is solved initially to compute optimal dispatch orders and the integer solutions are used in the continuous variable sub-problem to calculate actual arrival times of trains. Such a decomposition approach can be extended to rescheduling when dispatch orders of conflict trains can be pre-specified before computing decision variables.

Operational Control Approaches

Walker et al. (2005) proposed a new model for the combined scheduling of trains and crew; they extended it to a disruption recovery model. They defined disruptions as delays in arrival/departure of a train. The models are tested using a solver package and the resolving speed is quick enough to be used in real-time applications. Specific contribution toward application practice is the capability of the model to combine train and crew scheduling and a research initiative is to incorporate disruptions as delays to scheduled times. Train and crew scheduling is considered as a job scheduling problem and small time windows are created so that delay does not propagate further. A delay in small time window is considered as a new disruption to be resolved and the solutions are fed to further time windows to be solved as disruption recovery problems. Another rescheduling objective is to maximize the number of passengers using a train service (Diaz et al. 1999). A disruption is considered as a service delay and various alternate resolutions are computed using backtracking algorithms. A decision support system is developed that suggests alternate resolutions for the human dispatcher to choose from, based on a computed objective value. Sahin (1999) proposed a total delay minimization objective to resolve inter-train conflicts arising out of disturbance. Disturbance locations, time instants are specified, and trains that conflict are identified. Dispatch precedence is decided using a proposed ranking heuristic among two immediately conflicting trains detected one after the other. This heuristic holds merit as the precedence order decision is based on a look-ahead procedure to identify the number of potential conflicts that are consequences of a current resolution and delay duration of potential conflicts. Another significance of this heuristic is that delay location is specified, whereas most of the other approaches consider disruption as a delay in a time window, that is usually introduced from the first station in the operational domain. Narayanaswami (2009) developed a ranking based heuristic that decides dispatch priority based on a ranking procedure. Subsequently the ranking heuristic is developed in a multi-

Table 3 Railway Operations: Summary of Motivation, Objectives and Approaches

Operations	Motivation/Objectives			
	Scheduling	Approaches and Models		Rescheduling
Line planning	1. Number of Services 2. Operational costs	Optimization (Jovanovic and Harker 1991) Branch and Bound (Carey and Lockwood 1995, Lindner 2000) Simulation (Rangaraj et al. 2003, Gokhale 2004) Discrete Event Modeling (Dorfman and Medanic 2004) Dynamic Programming (Ko et al. 2008)	Optimization through simulation (Cheng 1996) Optimization (Bussieck 1998)	1. Energy Efficiency 2. Speed profile optimization 3. Capacity maximization 4. Reliability
Timetabling	1. Robustness 2. Delay resistance	Greedy algorithm (Cai and Goh 1994)	Genetic Algorithm and Tabu search (Gorman 1998) PESP (Liebchen and Möhring 2004) Integer Programming (Schöbel 2007) Real-time approach (Luèthi et al. 2007b) Heuristic (Burdett and Kozan 2008) ILP and Lagrangian (Cacchiani 2008) Dynamic Recoverable robustness (Cicerone et al. 2008) Stochastic optimization (Khan and Zhou 2009)	1. Reserve (buffer) distribution 2. Compensation, condensation area management 3. Integration of services 4. Disruption handling
Rolling stock management	1. Minimizing off balance (idle runs)	Optimization (Maròti 2006)	OPL model (Abbink et al. 2002) Optimization (Fioole et al. 2006) Optimization (Cacchiani et al. 2008) Rebalancing (Budai et al. 2009)	1. Off balance (idle runs) 2. Robustness 3. Recoverable 4. Punctuality
Crew scheduling	1. Optimizing crew utility 2. FCFS assignment 3. Freight crew	Decomposition (Caprara et al. 2007, Abbink et al. 2002) Branch and Bound (Freling et al. 2000) Fuzzy (Goncalves et al. 2000) Dantzig-Wolfe (Tatsushi et al. 2001) Ant-colony opt. (Wang et al. 2006) IP (Vaidyanathan et al. 2007)	Realtime (Walker et al. 2005) Column generation (Potthoff et al. 2010) Optimization (Veelenturf et al. 2012)	1. Minimize crew 2. Punctuality

Table 3 continued

Operations	Motivation/Objectives	
	Scheduling	Rescheduling
Policy decisions	1. Global service intention	1. Service improvement
	2. Integration of multi-modal traffic	2. Capacity increase
	3. Pricing of services	
	Approaches and Models	
	Priority analysis (Campbell 1996)	
	Competency assessment (Mahadevan 2007, Arslan 2009)	
	Realtime model (Wüest et al. 2008)	
	Heuristic (Leander and Lukaszewicz 2008)	

agents based framework. Cheng (1996) proposed a simulation based optimal traffic rescheduling by identifying the critical and non-critical paths in a schedule. A non-critical path is a schedule with time-table slacks and such non-critical paths are populated in a knowledge base. A simulation procedure identifies the optimal solution from the knowledge base by effectively utilizing the slacks for delay control. He further observed that real world train problems are ill structured and rule based heuristics are better applied in railway applications than traditional exact methods. In a later publication, Cheng (1998) developed a hybrid—network based and event driven simulation to resolve train conflicts. This approach is observed to be more efficient in dynamic environments. Perturbations in a single track schedule are handled as a discrete event based simulation and a local feedback based travel advance strategy is proposed by Dorfman and Medanic (2004). Other approximate conflict resolutions in railway applications are toward constraint propagation (Chiu et al. 1996), object oriented heuristic (Isaai and Singh 2000), heuristic approach at junctions (Ho and Yeung 2001), genetic algorithm based approaches (Chang and Chung 2005), (Nachtigall and Voget 1996), (Gorman 1998) and prototype based decision support system (Makkinga and Metselaar 1998).

Summary of Insights

Certain rescheduling issues that deserve more attention, other than formal optimization models to reschedule disrupted traffic and in real-time are identified. Rescheduling is directed towards recovering the original schedule. The primary arduous part in rescheduling is to check feasibility of whether a resource can be allocated to a task demand, without conflicting with other demands and deadlocks. It is possible to achieve feasibility, only if schedule slacks are available. Theoretically a schedule with zero slack can be developed even in sparse networks. Hence it is conclusive that rescheduling efficiency and feasibility are determined by amount and distribution of slack in a schedule. It is imperative that dependency between schedule slacks and rescheduling feasibility is considered as slack distribution in railway practice. However a research perspective in this context will lead to pertinacious measures of schedule reliability and capacity utilization. Some specific concepts to explore are (i) slacks to be formally represented as constraints in a rescheduling model, (ii) reliability study under slack constraints, and (iii) minimum slack estimate as threshold for rescheduling failure.

Conclusive Remarks

We present the broad set of literature on railway operations comprehensively in two tables. Table 3 summarizes motivation and objectives of scheduling and rescheduling of

Table 4 Railway research publications

Operations	Type of publications		
	PhD Theses	Technical notes/working papers	Journals
Line Planning	Bussieck (1998)—TU, Braunschwig Goossens (2004)—UM, Maastricht		TRB (1989)
			JORS (1995)
			Annals. OR (1998)
			CinR (2008)
Timetabling	Cacchiani (2008), Università di Bologna	ARRIVAL (2007)	TRB (1999)
			OR (2002)
			COR (2007)
			OR Spectrum (2008)
			ITS (2009)
			PT (2009)
Rolling stock management	Maròti (2006)—TU, Eindhoven	ERIM (2002, 2009)	J. Scheduling (1997) EJOR (2006)
Crew scheduling	Laplagne (2008)—University of Leeds Potthoff (2010)—Erasmus University, Rotterdam	IIMA (1986)	J.MP (1997)
		ERIM (2000, 2004)	PT (2001)
		EMS (2001)	COR (2005)
		Un.K (2002)	IBM – JRD (2007)
		Ec.Inst. (2004)	Interfaces (2011)
		EI (2010)	TRC (2012)
Policy decisions	Campbell (1996)—University of Pennsylvania, Philadelphia	RE (2008)	TRA (1980)
			IEEE-VT (1990, 1995)
			TS (1991)
			ERS (1997)
			DKE (1998)
			PT (2009) Transportation (2009)
Transportation	Lindner (2000)—TU, Braunschwig Gatto (2007)—ETH, Zurich D'Ariano (2008)—TRAIL Research School, Netherlands Luèthi (2009)—ETH, Zurich Narayanaswami (2009)—IIT, Bombay	Bilkent (1999)	COR (1994, 1996)
		Springer (1996, 2005, 2007, 2009-2)	TRB (1995, 1996, 1999, 2004, 2005, 2007)
		ERIM (2005)	SPA (1996)
		ARRIVAL (2009)	IEE—EPS (1996)
			CiI (1998)
			CiR (1998)
			IEEE-MSC (2000)
			IEE-EPA (2001)
			IEEE-ITS (2001)
			IEEE-VT (2001)
			TRA (2002)
			IST (2005)
			MMOR (2005)
			J. AMDS (2006)
			APJOR (2007)
			LSM (2007)
			Networks (2007)
			OR & MS (2007)
			EJOR (2008)
			NSE (2009-2)
			TRC (2009)

railway operations and the chronologically ordered research approaches published in literature. This table has two columns: Column 1 lists the railway scheduling and rescheduling operations that are covered in this paper. Column 2 lists the motives and objectives of each operation; this column is further sub-classified into three sub-columns. Leftmost sub-column lists scheduling objectives and rightmost sub-column lists rescheduling objectives. The sub-column at the center presents the models and approaches, along with literature cited in the order of their year of publishing; ones on the left correspond to scheduling and ones on the right correspond to rescheduling.

Table 4 reports chronologically ordered publications, as PhD theses, technical notes and journals on the different railway operations that we covered in this paper.

A spectrum of railway operational problems from research literature is presented. Methodological details of some of them and overview of many others are discussed. The focus of our paper is on scheduling and rescheduling of railway operations. An attempt is made to present literature classified based on operational problems and solution methodologies. Specific problem variations, that hold research potential and practical importance under each operational category are identified and reported. Several observations toward state of the art in strategic, tactical, operational control and real-time control of railway operations are reported. The article is concluded with a comment that the issue of rescheduling is valid even at strategic and tactical operations and a high level of rationalization in strategic operations modeling is a requisite to effective operational and real-time control operations. Specifically there is a dearth of formulating needs, models and approaches in rescheduling at strategic levels, and at real-time when disturbances are known just-in time. Further the interdependence between scheduling-rescheduling and the impact of one on the effectiveness of the other have to be studied more methodically.

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References

- Abbink, E., M. Fischetti, L. Kroon, G. Timmer, and M. Vromans. 2004. Reinventing Crew Scheduling at Netherlands Railways. ERIM Report Series Research in Management ERS-2004-046-LIS, Erasmus Research Institute of Management, Rotterdam.
- Abbink, E., B. Van den Berg, L. Kroon, and M. Salomon. 2002. Allocation of railway rolling stock for passenger trains. ERIM Report Series ERS-2002-47-LIS, Erasmus Research Institute of Management, Rotterdam.
- Abbink, E., J. van Wout, and D. Huisman. 2007. Solving large scale crew scheduling problems by using iterative partitioning. In *7th Workshop on Algorithmic Approaches for Transportation Modeling, Optimization, and Systems*, 96–106.
- Albrecht, T. 2009. The influence of anticipating train driving on the dispatching process in railway conflict situations. *Networks and Spatial Economics* 9(1): 85–101.
- Arslan, T. 2009. A hybrid model of fuzzy and AHP for handling public assessments on transportation projects. *Transportation* 36(1): 97–112.
- Assad, A.A. 1980. Models for rail transportation. *Transportation Research Part A: Policy and Practice* 14(3): 205–220.
- Beder, D.M., A. Romanovsky, B. Randell, C.R. Snow, and R.J. Stroud. 2000. An application of fault tolerance patterns and coordinated atomic actions to problem in railway scheduling. In *ACM SIGOPS Operating Systems Review*, 34, 21–31.
- Bergkvist, M., P. Davidsson, J.A. Persson and L. Ramstedt., 2005. A hybrid micro-simulator for determining the effects of governmental control policies on transport chains. In *Joint AAMAS Workshop on Multi-Agent and Multi-Agent-Based Simulation*, 236–247.
- Blum, J., and A. Eskandarian. 2002. Enhancing intelligent agent collaboration for flow optimization of railroad traffic. *Transportation Research Part A: Policy and Practices* 36(10): 919–930.
- Bojovic, N. and M. Milenkovic, M. 2010. Train driver rostering optimization. In *8th International Symposium on Intelligent Systems and Informatics (SISY)*, 501–504, Subotica.
- Brucker, P., S. Heitmann, and S. Knust. 2005. *Scheduling railway traffic at a construction site*. Berlin: Springer.
- Budai, G., G. Maróti, R. Dekker, D. Huisman, and L. Kroon. 2009. Rescheduling in passenger railways: The rolling stock rebalancing problem. *Journal of Scheduling* Volume 13(3): 281–297.
- Burdett, R., and E. Kozan. 2008. A sequencing approach for creating new train timetables. *OR Spectrum* 32(1): 163–193. doi: 10.1007/s00291-008-0143-6.
- Bussieck, M. 1998. Optimal lines in public rail transport. PhD thesis, Technische Universität Braunschweig, Braunschweig.
- Cacchiani, V. 2008. Models and algorithms for combinatorial optimization problems arising in railway applications. PhD thesis, Università di Bologna.
- Cacchiani, V., A. Caprara, L. Galli, L. Kroon, and G. Maróti. 2008. Recoverable robustness for railway rolling stock planning. In *ATMOS 2008—8th Workshop on Algorithmic Approaches for Transportation Modeling, Optimization, and Systems*, ed. Fischetti, M. and Widmayer, P. Dagstuhl: Schloss Dagstuhl—Leibniz-Zentrum fuer Informatik.
- Cai, X., and C.J. Goh. 1994. A fast heuristic for the train scheduling problem. *Computers & Operations Research* 21(5): 499–510.
- Caimi, G., F. Fuchsberger, D. Burkolter, T. Herrmann, R. Wüst, and S. Roos. 2009. Conflict-free train scheduling in a compensation zone exploiting the speed profile. In *ISRROR Proceedings*, Seoul.
- Campbell, K.C. 1996. Booking and revenue management for rail intermodal services. PhD thesis, University of Pennsylvania, Philadelphia.
- Caprara, A., M. Fischetti, and P. Toth. 2002. Modeling and solving the timetabling problem. *Operations Research* 50(5): 851–861.
- Caprara, A., M. Fischetti, P. Toth, D. Vigo, and P.L. Guida. 1997. Algorithms for railway crew management. *Journal of the mathematical programming: Series A and B—Special issue: The 16th international symposium on mathematical programming*, 79(1–3).
- Caprara, A., L.G. Kroon, M. Monaci, M. Peeters, and P. Toth. 2007. Passenger railway optimization, Chap. 3. In *Handbooks in operations research and management science*, ed. Barnhart, C. and Laporte, G. (vol. 14, Chap. 3), 129–187. Amsterdam: Elsevier.
- Carey, M., and D. Lockwood. 1995. A model, algorithms and strategy for train pathing. *Journal of the Operations Research Society* 46: 988–1005.

- Chang, C.S., C.S. Chua, H.B. Quek, X.Y. Xu, and S.L. Ho. 1998. Development of Train Movement Simulator for analysis and Optimization of railway signalling systems. In *International Conference on Developments in Mass Transit Systems* 543: 243–248.
- Chang, C.S., and B.S. Thia. 1996. Online rescheduling of mass rapid transit systems: Fuzzy expert system approach. In *IEEE Proceedings: Electrical Power Applications* 143: 307–316.
- Chang, S.C., and Y.-C. Chung. 2005. From timetabling to train regulation—a new train operation model. *Information and Software Technology* 47: 575–585.
- Chen, B., H.H. Cheng, and J. Palen. 2009. Integrating mobile agent technology with multi-agent systems for distributed traffic detection and management systems. *Transportation Research Part C: Emerging Technologies* 17(1): 1–10.
- Cheng, Y. 1996. Optimal train traffic rescheduling simulation by a knowledge-based system combined with critical path method. *Simulation Practice and Theory* 4: 399–413.
- Cheng, Y. 1998. Hybrid simulation for resolving resource conflicts in train traffic rescheduling. *Computers in Industry* 35(3): 233–246.
- Chiang, T.-W., H.-Y. Hau, H.-M. Chiang, S.-Y. Ko, and C.-H. Hsieh. 1998. Knowledge based system for railway scheduling. *Data & Knowledge Engineering* 27(3): 289–312.
- Chiu, C.K., C.M. Chou, J.H.M. Lee, H.F. Leung, and Y.W. Leung. 1996. *A constraint-based interactive train rescheduling tool*, 104–118. Berlin: Springer.
- Cicerone, S., G.D. Stefano, M. Schachtebeck, and A. Schöbel. 2008. Dynamic algorithms for recoverable robustness problems. In *ATMOS 2008—8th Workshop on Algorithmic Approaches for Transportation Modeling, Optimization, and Systems*, ed. Fischetti, M. and Widmayer, P. Dagstuhl: Schloss Dagstuhl—Leibniz-Zentrum fuer Informatik.
- Cordeau, J.F., P. Toth, and D. Vigo. 1998. A survey of optimization models for train routing and scheduling. *Transportation Science* 32(4): 380–404.
- D'Ariano, A. 2008. Improving real-time train dispatching: Models, algorithms and applications. PhD thesis, TRAIL Research School.
- D'Ariano, A., and M. Pranzo. 2009. An advanced real-time train dispatching system for minimizing the propagation of delays in a dispatching area under severe disturbances. *Networks and Spatial Economics* 9(1): 63–84.
- Davidsson, P., L. Henesey, L. Ramstedt, J. Tornquist, and F. Wernstedt. 2005. An analysis of agent-based approaches to transport logistics. *Transportation Research Part C: Emerging Technologies* 13(4): 255–271.
- Diaz, A. 2001. Train scheduling. In *Fifth International Conference on Autonomous Agents*, Montreal, Canada.
- Diaz, A., M.O. Gonzalez, and P. Gonzalez-Torre. 1999. On-line timetable re-scheduling in regional train services. *Transportation Research Part B: Methodological* 33(6): 387–398.
- Dorfman, M.J., and J. Medanic. 2004. Scheduling trains on a railway network using a discrete event model of railway traffic. *Transportation Research Part B: Methodological* 38(1): 81–98.
- Fioole, P.-J., L. Kroon, G. Maroti, and A. Schrijver. 2006. A rolling stock circulation model for combining and splitting of passenger trains. *European Journal of Operational Research* 174(2): 1281–1297.
- Fischetti, M., D. Salvagnin, and A. Zanette. 2007. Fast approaches to improve the robustness of a railway timetable. Technical report, ARRIVAL project.
- Frede, L., M. Müller-Hannemann, and M. Schnee. 2008. Efficient on-trip timetable information in the presence of delays. In *ATMOS 2008—8th Workshop on Algorithmic Approaches for Transportation Modeling, Optimization, and Systems*, ed. Fischetti, M. and Widmayer, P. Dagstuhl: Schloss Dagstuhl—Leibniz-Zentrum fuer Informatik.
- Freling, R., R. Lentink, and M.A. Odijk. 2000. Scheduling train crews: A case study for the Dutch railways. Technical 17, Econometric Institute, Rotterdam.
- Gatto, M., R. Jacob, L. Peeters, and P. Widmayer. 2009. *On-Line delay management on a single train line. Lecture Notes in Computer Science*. Heidelberg: Springer Berlin.
- Gatto, M.J. 2007. On the impact of uncertainty on some optimization problems: Combinatorial aspects of delay management and robust online scheduling. PhD thesis, Swiss Federal Institute of Technology, ETH Zurich.
- Gely, L., D. Feillee, and G. Dessagne. 2009. A co-operative framework between optimization and simulation to address on-line re-scheduling problems. Technical report, ARRIVAL project.
- Ghoseiri, K. and F. Morshedsolouk. 2006. ACS-TS: Train scheduling using ant colony system. *Journal of Applied Mathematics and Decision Sciences*, Article ID 95060, 28. doi:10.1155/JAMDS/2006/95060.
- Godwin, T., R. Gopalan, and T.T. Narendran. 2007a. A heuristic for routing and scheduling freight trains in a passenger rail network. *International Journal of Logistics Systems and Management* 3: 101–133.
- Godwin, T., R. Gopalan, and T.T. Narendran. 2007b. Freight train routing and scheduling in a passenger rail network: Computational complexity and the stepwise dispatching heuristic. *Asia Pacific Journal of Operational research* 24(4): 499–533.
- Gokhale, R. 2004. Simulation studies on railway line capacity. Master's thesis, IEOR, Indian Institute of Technology, Mumbai.
- Goncalves, R., R. Lagrimante, and F. Gomide. 2000. Improving railway crew scheduling quality: A hybrid approach. In *19th International Conference of the North American Fuzzy Information Processing Society*, 378–382.
- Goossens, J.-W. 2004. Models and algorithms for railway line planning problems. PhD thesis, Universiteit Maastricht.
- Gorman, F.M. 1998. An application of genetic and tabu searches to the freight railroad operating plan problem. *Annals of Operations Research* 78(19): 51–69.
- Heilporn, G., L. De Giovanni, and M. Labbè. 2008. Optimization models for the single delay management problem in public transportation. *European Journal of Operational Research* 189(3): 762–774.
- Hellstrom, P., B. Sandblad, I. Frej, and A. Gideon. 1998. An evaluation of algorithms and systems for computer aided train dispatching. *Proceedings of COMPTAIL 98*, Lisbon.
- Higgins, A., E. Kozan, and L. Ferreira. 1996. Optimal scheduling of trains on a single line track. *Transportation Research Part B: Methodological* 30(2): 147–161.
- Ho, T.K. and T.H. Yeung. 2001. Railway junction traffic control by heuristic methods. In *IEEE Proceedings—Electrical Power Applications*, vol. 148, 77–84.
- Huisman, D., L.G. Kroon, R.M. Lentink, and M.J.C.M. Vromans. 2005. Operations research in passenger railway transportation. Technical report, Erasmus Research Institute of Management (ERIM), Rotterdam School of Management.
- Ingolitto, L., F. Barber, P. Tormos, A. Lova, M. Salido, and M. Abril. 2004. An efficient method to schedule new trains on a heavily loaded railway network. In *Proceedings of Advances in Artificial Intelligence—IBERAMIA 2004*, 164–173.
- Isaai, M., and M. Singh. 2000. An object-oriented constraint based heuristic for a class of passenger-train scheduling problems. *IEEE Transactions on Man, Systems and Cybernetics Part C: Applications and Reviews* 30(1): 12–21.
- Iyer, R.V., and S. Ghosh. 1995. DARYN: A distributed decision making algorithm for railway networks: Modeling and simulation. *IEEE Transactions on Vehicular Technology* 44: 180–191.

- Jespersen-Groth, J., D. Potthoff, J. Clausen, D. Huisman, L.G. Kroon, G. Maroti, and M.N. Nielsen. 2009. *Disruption management in passenger railway transportation. robust and online large-scale optimization*. Berlin: LNCS, Springer.
- Jovanovic, D., and P. Harker. 1991. Tactical scheduling of rail operations: SCAN I system. *Transportation Sciences* 25(1): 46–64.
- Jütte, S., M. Albers, U.W. Thonemann, and K. Haase. 2011. Optimizing railway crew scheduling at DB Schenker. *Interfaces* 41(2): 109–122.
- Kawakami, T. 1999. Dynamic power saving strategy and data system for future shinkansen traffic control. In *ISADS'99: Proceedings of the The Fourth International Symposium on Autonomous Decentralized Systems*. IEEE Computer Society, 18, Washington.
- Keaton, M.H. 1989. Designing optimal railroad operating plans: lagrangian relaxation and heuristic approaches. *Transportation research Part B: Methodological*, 23 B(6): 415–431.
- Khan, M.B., and X. Zhou. 2009. Stochastic optimization model and solution algorithm for robust double-track train-timetabling problem. *IEEE Transactions on Intelligent Transportation Systems* 11(1): 81–89. doi:10.1109/TITS.2009.2030588.
- Kizilisik, O.B. 1999. Predictive and reactive scheduling. <http://www.ie.bilkent.edu.tr/~ie672/docs/present/kizilisik.ps>.
- Ko, H., T. Koseki, and M. Miyatake. 2008. Application of dynamic programming to optimization of running profile of a train. *Computers in Railways IX*.
- Kraay, D., and P. Harker. 1995. Real-time scheduling of freight railroads. *Transportation Research Part B: Methodological* 29B(3): 213–229.
- Kroon, L. and M. Fischetti. 2000. Crew scheduling for Netherlands railways: “destination: customer”. Research Paper ERS-2000-56-LIS, Erasmus Research Institute of Management (ERIM).
- Laplagne, I.E. 2008. Train driver scheduling with windows of relief opportunities. PhD thesis, University of Leeds.
- Leander, P. and P. Lukaszewicz. 2008. EETROP: Energy efficient train operation-state of the art of train ECO-operation. Technical Report XX, Railenergy WP 2.3.
- Lee, T.S., and S. Ghosh. 2001. Stability of RYNSORD: A decentralised algorithm for railway networks under perturbation. *IEEE Transactions on Vehicular Technology* 50(1): 287–301.
- Li, J.-Q., P.B. Mirchandani, and D. Borenstein. 2007. The vehicle rescheduling problem: Model and algorithms. *Networks* 50(3): 211–229.
- Liebchen, C. and R.H. Möhring. 2004. The modeling power of the periodic event scheduling problem: Railway timetables and beyond. In *ATMOS*, 3–40.
- Liebchen, C., M. Schachtebeck, A. Schöbel, S. Stiller, and A. Prigge. 2007. Computing delay resistant railway timetables. *Computers & Operations Research* 37(5): 857–868.
- Liebchen, C., and S. Stiller. 2009. Delay resistant timetabling. *Public transport* 1(1): 55–72.
- Lin, H.-C. and C.-C. Hsu. 1994. An interactive train scheduling workbench based on artificial intelligence. In *Proceedings of the International Conference on Tools with Artificial Intelligence—ICTAI*, 42–48.
- Lindner, T. 2000. Train schedule optimization in public rail transport. PhD thesis, Technischen Universität Braunschweig, Braunschweig.
- Lindner, T., and U. Zimmermann. 2005. Cost optimal periodic train scheduling. *Mathematical Methods of Operations Research* 62(15): 281–295.
- Luèthi, M. 2009. Improving the efficiency of heavily used railway networks through integrated real-time rescheduling. PhD thesis, Swiss Federal Institute of Technology, ETH Zurich.
- Luèthi, M., G. Medeossi, and A. Nash. 2007a. Evaluation of an integrated real-time rescheduling and train control system for heavily used areas. In *International Seminar on Railway Operations Modelling and Analysis (IAROR) Conference*, Hannover.
- Luèthi, M., A. Nash, U. Weidmann, F. Laube, and R. Wüst. 2007b. Increasing railway capacity and reliability through integrated real-time rescheduling. In *11th World Conference on transportation research*, Berkeley.
- Mahadevan, V. 2007. Describing and evaluating train services on the swiss railway network from a new perspective. In *7th (STRC) Swiss Transport Research Conference*, Monte Verita, Ascona.
- Makkinga, F. and S. Metselaar. 1998. Automatic conflict detection and advanced decision support for optimal usage of railway infrastructure prototyping and test results. In *Computers in railways-VI*, vol. 37. The Built Environment.
- Maroti, G. 2006. Operations research models for railway rolling stock planning. PhD thesis, Technische Universiteit, Eindhoven, Amsterdam.
- Mazzarello, M., and E. Ottaviani. 2007. A traffic management system for real-time traffic optimisation in railways. *Transportation Research Part B: Methodological* 41(2): 246–274.
- Michaelis, M., and A. Schöbel. 2009. Integrating line planning, timetabling, and vehicle scheduling: A customer-oriented heuristic. *Public Transport* 1(3): 211–232.
- Möller, J. 2002. Seminar on algorithms and models for railway optimization crew scheduling. University of Konstanz, Ilerseminar.
- Müller-Hannemann, M., F. Schulz, D. Wagner, and C.D. Zaroliagis. 2004. Timetable information: Models and algorithms. In *4th Workshop on Algorithmic methods and models for Optimization of Railways—ATMOS*, 67–90.
- Nachtigall, K., and S. Voget. 1996. A genetic algorithm approach to periodic railway synchronization. *Computers & Operations Research* 23(5): 453–463.
- Narayanaswami, S. 2009. Dynamic railway rescheduling using intelligent agents. PhD thesis, Indian Institute of Technology Bombay, Powai, Mumbai.
- Nielsen, L. and G. Maroti. 2009. Disruption management of rolling stock in passenger railway transportation. Research Paper ERS-2009-046-LIS, Erasmus Research Institute of Management (ERIM).
- Parkes, D. and L. Ungar. 2001. An auction based method for decentralised train scheduling. In *Fifth International Conference on Autonomous Agents*, Montreal.
- Ping, L., N. Axin, J. Limin, and W. Fuzhang. 2001. Study on intelligent train dispatching. *IEEE Intelligent Transportation Systems*, 949–953.
- Potthoff, D. 2010. Railway crew rescheduling: Novel approaches and extensions. PhD thesis, Erasmus University, Rotterdam.
- Potthoff, D., D. Huisman, and G. Desaulniers. 2010. Column generation with dynamic duty selection for railway crew rescheduling. *EI reprint reeks EI-1561*.
- Raghuram, G., V. Venkata Rao, and S. Jahar. 1986. Crew scheduling in Indian railways: A case study of Vadodara division passenger train crews. IIMA Working Papers WP1986-11-01-00718, Indian Institute of Management Ahmedabad, Research and Publication Department.
- Rangaraj, N., A. Ranade, K. Moudgalya, C. Konda, M. Johri, and R. Naik. 2003. Simulator for railway line capacity planning. In *Operational Research and its Applications: Recent Trends, Proceedings of the APORS*.
- Sahin, I. 1999. Railway traffic control and train scheduling based on inter-train conflict management. *Transportation Research Part B: Methodological* 1(33): 511–534.
- Salim, V., and X. Cai. 1997. A genetic algorithm for railway scheduling with environmental considerations. *Environmental modelling and Software* 12(4): 309–310.

- Schachtebeck, M. and A. Schöbel. 2008. IP-based techniques for delay management with priority decisions. In *8th Workshop on Algorithmic methods and models for Optimization of Railways—ATMOS*.
- Schöbel, A. 2007. Integer programming approaches for solving the delay management problem. In *Algorithmic methods for railway optimization*, 145–170. Heidelberg: LNCS, Springer.
- Taïeb, Mellouli 2001. A network flow approach to crew scheduling based on an analogy to an aircraft/train maintenance routing problem. *Lecture Notes in Economics and Mathematical Systems*, 505(1): 91–120.
- Tatsushi, N., M. Yuki, and I. Masahiro. 2001. Column generation with dual inequalities for railway crew scheduling problems. *Public Transport* 3(1): 1–18.
- Tornquist, J. 2006. Computer based decision support for railway traffic scheduling and dispatching: A review of models and algorithms. In *5th Workshop on Algorithmic methods and models for Optimization of Railways—ATMOS 2005*.
- Tornquist, J. 2007. Railway traffic disturbance management: An experimental analysis of disturbance complexity, management objectives and limitations in planning horizon. In *Transportation Research Part A: Policy and Practice*, 41: 249–266.
- Tornquist, J. and P. Davidsson. 2002. A multi-agent system approach to train delay handling. In *Workshop on Agent Technologies in Logistics*, 50–54, Lyon.
- Tornquist, J. and J.A. Persson. 2005. A n-tracked railway traffic during disturbances: Theoretical and practical implications. *Transportation Research Part B: Methodological*, 41(3): 342–362. doi:[10.1016/j.trb.2006.06.001](https://doi.org/10.1016/j.trb.2006.06.001).
- Vaidyanathan, B., K.C. Jha, and R.K. Ahuja. 2007. Multicommodity network flow approach to the railroad crew-scheduling problem. *IBM Journal of Research and Development* 51(3.4): 325–344.
- Veelenturf, L., D. Potthoff, D. Huisman, and L. Kroon. 2012. Railway crew rescheduling with retiming. *Transportation Research. Part C. Emerging Technologies* 20(1): 95–110.
- Vernazza, G., and R. Zunino. 1990. A distributed intelligent methodology for railway traffic control. *IEEE Transactions on Vehicular Technology* 39(3): 263–270.
- Walker, C.G., J.N. Snowdon, and D.M. Ryan. 2005. Simultaneous disruption recovery of a train timetable and crew roster in real time. *Computers & Operations Research* 32: 2077–2094.
- Wang, F., H. Wang, and J. Shen. 2006. Modeling and solving for railway crew scheduling problem. In *Intelligent Control and Automation, 2006. WCICA 2006. The Sixth World Congress on*, 2: 8621–8624.
- Wüest, R., F. Laube, S. Roos, and G. Caimi. 2008. Sustainable global service intention as objective for controlling railway network operations in real time. In *Proceedings of the 8th World Congress of Railway Research (WCRR)*, Seoul, Korea.
- Yoko, T. and T. Norio. 2005. Robustness indices for train rescheduling. In *1st International Seminar on Railway Operations Modelling and Analysis*, Delft.